

Problem

A 60-W incandescent bulb operates at 120 V. How many electrons and coulombs flow through the bulb in one day?

Step-by-step solution

Step 1 of 3

Consider the following data:

A 60-W incandescent bulb operates at 120 V .

$$p = 60\text{-W}$$

$$V_o = 120\text{ V}$$

Calculate the value of current by using following formula.

$$p = V_o i$$

$$i = \frac{p}{V_o}$$

Substitute the values 60-W for p and 120 V for V_o to calculate the current i .

$$\begin{aligned} i &= \frac{60\text{-W}}{120\text{ V}} \\ &= 0.5\text{ A} \end{aligned}$$

Step 2 of 3

Electric current is the time rate of change of charge.

$$i \triangleq \frac{dq}{dt}$$

For $t = 1\text{ second}$, $i \triangleq dq$

$$\begin{aligned} q &= i \\ &= 0.5\text{ C} \end{aligned}$$

Calculate the coulombs flow through the bulb in one day as follows.

$$\begin{aligned} \text{Coulombs flow through the bulb in one day} &= (0.5\text{ C})(1\text{ day}) \\ &= (0.5\text{ C})(24\text{ h})(60\text{ min})(60\text{ s}) \\ &= (0.5\text{ C})(24)(60)(60) \\ &= 43200\text{ C} \end{aligned}$$

Therefore, the coulombs flow through the bulb in one day is 43200C .

Step 3 of 3

Calculate the electrons flow through the bulb in one day as follows.

Consider 1 C of charge contains 6.24×10^{18} electrons.

$$\begin{aligned}\text{Electrons flow through the bulb in one day} &= (43200\text{ C}) (6.24 \times 10^{18} \text{ electrons}) \\ &= 269568 \times 10^{18} \text{ electrons} \\ &= 2.696 \times 10^{23} \text{ electrons}\end{aligned}$$

Therefore, the electrons flow through the bulb in one day is 2.696×10^{23} electrons.